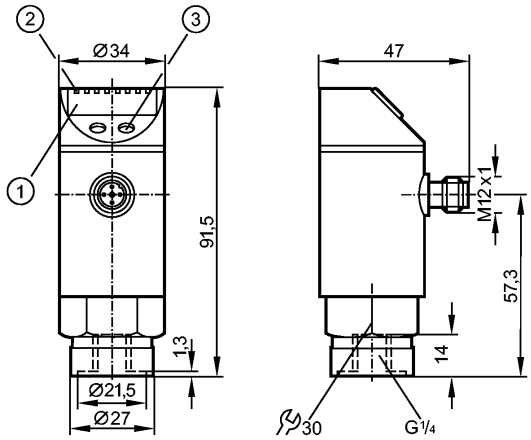
efectorsoo
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PN2020

PN-400-SBR14-MFRKG/US/ /V

Pressure sensors



1: 4-digit alphanumeric display
2: LEDs (display unit / switching status)
3: Programming button

Made in Germany

CE cULus LISTED IO-Link

Product characteristics

- Combined pressure sensor
- Connector
- Process connection: G¼ I
- Communication interface: IO-Link 1.0 (COM2 slave, 38.4 kBaud)
- Zero and span adjustable
- Function programmable
- 2 outputs
  OUT1 = switching output
  OUT2 = switching output or analogue output
- 4-digit alphanumeric display
- Measuring range: 0...400 bar / 0...5800 psi / 0...40 MPa

Application

Application		Type of pressure: relative pressure Liquids and gases Use in gases at pressures > 25 bar only after contacting the manufacturer ifm		
Pressure rating		600 bar	8700 psi	60 MPa
Bursting pressure min.		1000 bar	14500 psi	100 MPa
Medium temperature	[°C]	-25...80		

Electrical data

Electrical design		DC PNP/NPN
Operating voltage	[V]	18...32 DC [1])
Current consumption	[mA]	< 35
Insulation resistance	[MΩ]	> 100 (500 V DC)
Protection class		III
Reverse polarity protection		yes

Outputs

Output	2 outputs OUT1 = switching output OUT2 = switching output or analogue output

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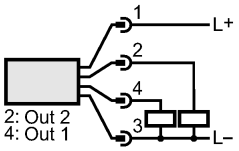
PN-400-SBR14-MFRKG/US/ /V

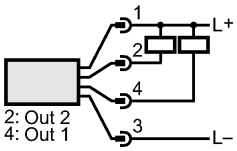
Pressure sensors

Output function	2 x normally open / closed programmable or 1 x normally open / closed programmable + 1 x analogue (4...20 mA / 0...10 V; scaleable 1:4)		
Current rating [mA]	2 x 250		
Voltage drop [V]	< 2		
Short-circuit protection	pulsed		
Overload protection	yes		
Switching frequency [Hz]	≤ 500		
Analogue output	4...20 mA / 0...10 V		
Max. load [Ω]	4...20 mA: max. (Ub - 10 V) x 50 / 0...10 V: min. 2000		
Measuring / setting range			
Measuring range	0...400 bar	0...5800 psi	0...40 MPa
Setting range			
Set point, SP	4...400 bar	60...5800 psi	0.4...40.0 MPa
Reset point, rP	2...398 bar	30...5770 psi	0.2...39.8 MPa
Analogue start point, ASP	0...300 bar	0...4350 psi	0.0...30.0 MPa
Analogue end point, AEP	100...400 bar	1450...5800 psi	10.0...40.0 MPa
in steps of	1 bar	10 psi	0.1 MPa
Factory setting	SP1 = 100 bar; rP1 = 92 bar SP2 = 300 bar; rP2 = 292 bar ASP = 0 bar; AEP = 400 bar		
Accuracy / deviations			
Accuracy / deviations (in % of the span) Turn down 1:1			
Switch point accuracy	< ± 0.4		
Characteristics deviation *)	< ± 0.25 (BFSL) / < ± 0.5 (LS)		
Hysteresis	< ± 0.1		
Repeatability **)	< ± 0.1		
Long-term stability ***)	< ± 0.1		
Temperature coefficients (TEMPCO) in the temperature range 0...80° C (in % of the span per 10 K)			
Greatest TEMPCO of the zero point	< ± 0.2		
Greatest TEMPCO of the span	< ± 0.2		
Reaction times			
Power-on delay time [s]	0.3		
Min. response time switching output [ms]	1.5		
Damping for the switching output (dAP) [s]	0; 0.01...4.00		
Damping for the analogue output (dAA) [s]	0; 0.01...4.00		
Response time analogue output [ms]	3		
Integrated watchdog	yes		
Software / programming			
Programming options	hysteresis / window function; N.O. / N.C; output polarity; current / voltage outputs; damping; calibration of displayed values; display can be rotated / deactivated; display unit		
Interfaces			
IO-Link Device			
Transfer type	COM2 (38.4 kBaud)		

IO-Link revision		1.0
IO-Link Device ID		56 d / 00 00 38 h
Profiles		no profile
SIO mode		yes
Required master port type		A
Process data analogue		1
Process data binary		2
Min. process cycle time	[ms]	2.3
Environment		
Ambient temperature	[°C]	-25...80
Storage temperature	[°C]	-40...100
Protection		IP 67
Tests / approvals		
EMC		EN 61000-4-2 ESD: 4 kV CD / 8 kV AD EN 61000-4-3 HF radiated: 10 V/m EN 61000-4-4 Burst: 2 kV EN 61000-4-5 Surge: 0.5/1 kV EN 61000-4-6 HF conducted: 10 V
Shock resistance		DIN IEC 68-2-27: 50 g (11 ms)
Vibration resistance		DIN IEC 68-2-6: 20 g (10...2000 Hz)
MTTF	[Years]	131
Mechanical data		
Process connection		G¼ I
Materials (wetted parts)		stainless steel (303S22); ceramics; FPM (Viton)
Housing materials		stainless steel (304S15); stainless steel 316L / 1.4404; PC (Makrolon); PBT (Pocan); PEI; FPM (Viton)
Switching cycles min.		100 million
Weight	[kg]	0.267
Displays / operating elements		
Display		Display unit 3 x LED green Switching status 2 x LED yellow Function display 4-digit alphanumeric display Measured values 4-digit alphanumeric display
Electrical connection		
Connection		M12 connector; gold-plated contacts
Wiring		

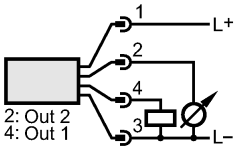
Programming of the output function
(OUT1 / OUT2):
Hno = hysteresis / normally open
Hnc = hysteresis / normally closed
Fno = window function / normally open
Fnc = window function / normally closed
Complementary outputs:
output 1: = Hno, output 2: = Hnc
(with the same SP / rP)

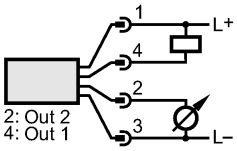




Programming of the analogue output (OUT2):
I = current output (4...20 mA)
U = voltage output (0...10 V)







Remarks

Remarks	[1]) to EN50178, SELV, PELV *) BFSL = Best Fit Straight Line / LS = Limit Value Setting **) with temperature fluctuations < 10 K ***) in % of the span per year
Pack quantity [piece]	1

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